Applied Economics

Multiple Linear Regression, Causality, and Inference

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Plan for this Lecture

1. The causal question: potential outcomes, ATE, ATT
2. Regression as a tool: what $\hat{\beta}$ measures and when it is causal
3. Why OLS fails: one formula, three sources of endogeneity
4. Inference: robust and clustered standard errors
5. Bridge: what IV, panel FE, DiD and RDD each assume instead of $\mathbb{E}[u_i \mid X_i] = 0$

The Multiple Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i, \qquad i = 1, \ldots, N$$

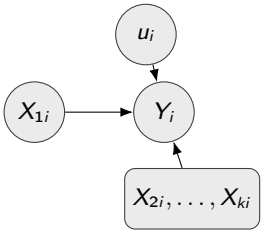
- i: unit of observation in the population.
- Y_i: **dependent variable** (outcome).
- X_{1i}: **treatment** variable, our main focus.
- β_1: coefficient for the treatment, **same** for all i.
- $X_{2i}, \ldots, X_{ki}$: **covariates** (controls), observed.
- β_{ki}: coefficient for the k-th control variable.
- u_i: **error term**: all *unobserved* determinants of Y_i.

- Correlation: a statistical **measure** describing the relation between treatment X_{1i} and outcome Y_i.
- Causality: a **claim** about the effect of X_{1i} on Y_i, holding the controls and u_i fixed.

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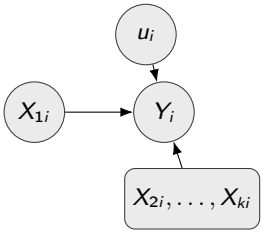


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Potential Outcomes

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i, \qquad i = 1, \ldots, N \qquad X_{1i} \in \{0, 1\}$$

- $Y_i(0) = \beta_0 + \beta_1 \cdot 0 + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$: outcome of i without treatment.
- $Y_i(1) = \beta_0 + \beta_1 \cdot 1 + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$: outcome of i with treatment.

Each observation belongs to either the treated or control group and has two **potential outcomes**:

	$X_{1i} = 1$ (treated)	$X_{1i} = 0$ (control)
Observed outcome	$Y_i = Y_i(1)$	$Y_i = Y_i(0)$
Missing counterfactual	$Y_i(0)$	$Y_i(1)$

Causal effect of X_{1i} on **individual** unit i: $Y_i(1) - Y_i(0) \quad \rightarrow \quad$ Cannot be computed.

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From Individual Effects to Group Comparisons

Average Treatment Effect, ATE

Compares mean outcomes of treated and controled units with same controls $\boldsymbol{X}_i = (X_{2i}, \ldots, X_{ki})$:

$$ATE = \underbrace{\mathbb{E}[Y_i(1) \mid X_{1i} = 1, \boldsymbol{X}_i]}_{\text{Treated}} - \underbrace{\mathbb{E}[Y_i(0) \mid X_{1i} = 0, \boldsymbol{X}_i]}_{\text{Control}}$$

Adding and subtracting the treated group's missing counterfactual...

$$\begin{aligned} ATE &= \mathbb{E}[Y_i(1) \mid X_{1i} = 1, \boldsymbol{X}_i] - \mathbb{E}[Y_i(0) \mid X_{1i} = 1, \boldsymbol{X}_i] + \mathbb{E}[Y_i(0) \mid X_{1i} = 1, \boldsymbol{X}_i] - \mathbb{E}[Y_i(0) \mid X_{1i} = 0, \boldsymbol{X}_i] \\ &= \underbrace{\mathbb{E}[Y_i(1) - Y_i(0) \mid X_{1i} = 1, \boldsymbol{X}_i]}_{\text{causal effect (on the treated)}} + \underbrace{\mathbb{E}[Y_i(0) \mid X_{1i} = 1, \boldsymbol{X}_i] - \mathbb{E}[Y_i(0) \mid X_{1i} = 0, \boldsymbol{X}_i]}_{\text{selection bias}} \end{aligned}$$

Selection Bias

- How the treated and control group outcomes would differ *even without treatment*.

→ Main purpose of causal inference: get rid of selection bias.

Bringing MLR and ATE Together

Plugging in the regression model: $Y_i(X_{1i}) = \beta_0 + \beta_1 \cdot X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i \qquad X_{1i} \in \{0, 1\}$

1. The causal effect simplifies to: $\mathbb{E}[Y_i(1) - Y_i(0) \mid X_{1i} = 1, \boldsymbol{X}_i] = \beta_1$
2. The selection bias $\mathbb{E}[Y_i(0) \mid X_{1i} = 1, \boldsymbol{X}_i] - \mathbb{E}[Y_i(0) \mid X_{1i} = 0, \boldsymbol{X}_i]$ splits into:
   - differences in controls: $\sum_{j=2}^{k} \beta_j \left(\mathbb{E}[X_{ji} \mid X_{1i} = 1, \boldsymbol{X}_i] - \mathbb{E}[X_{ji} \mid X_{1i} = 0, \boldsymbol{X}_i]\right) = \sum_{j=2}^{k} \beta_j \left(X_{ji} - X_{ji}\right) = 0$
   - differences in unobservables: $\mathbb{E}[u_i \mid X_{1i} = 1, \boldsymbol{X}_i] - \mathbb{E}[u_i \mid X_{1i} = 0, \boldsymbol{X}_i]$

ATE in a Multiple Linear Regression Model with Binary Treatment becomes:

$$ATE = \beta_1 + \underbrace{\mathbb{E}[u_i \mid X_{1i} = 1, \boldsymbol{X}_i] - \mathbb{E}[u_i \mid X_{1i} = 0, \boldsymbol{X}_i]}_{\text{selection bias}}$$

When does observed ATE represent true unobserved causal effect β_1?

When does observed ATE also include unobserved selection bias?

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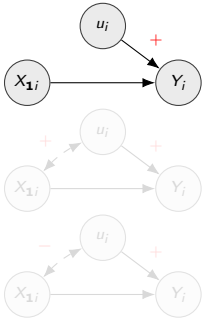
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Average Treatment Effect and Selection Bias

Selection Bias in MLR with Binary Treatment

$$\mathbb{E}[u_i \mid X_{1i} = 1, \boldsymbol{X}_i] - \mathbb{E}[u_i \mid X_{1i} = 0, \boldsymbol{X}_i] = \frac{\mathrm{Cov}(X_{1i}, u_i \mid \boldsymbol{X}_i)}{\mathrm{Var}(X_{1i} \mid \boldsymbol{X}_i)}$$

- **No bias**: $\mathrm{Corr}(X_{1i}, u_i \mid \boldsymbol{X}_i) = 0 \rightarrow$ observed $ATE =$ unobserved β_1.
- **+ bias**: $\mathrm{Corr}(X_{1i}, u_i \mid \boldsymbol{X}_i) > 0 \rightarrow$ observed $ATE >$ unobserved β_1.
- **− bias**: $\mathrm{Corr}(X_{1i}, u_i \mid \boldsymbol{X}_i) < 0 \rightarrow$ observed $ATE <$ unobserved β_1.

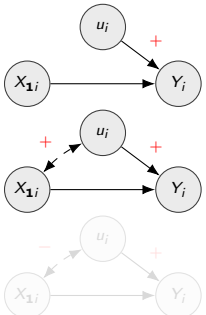


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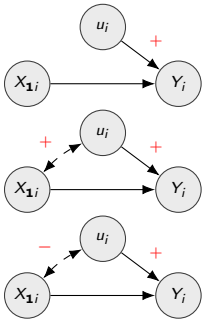


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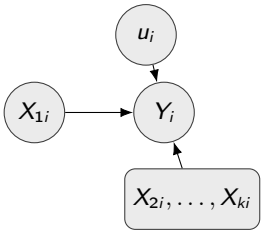
Why RCTs Work

Randomized control trial (RCT): treatment is assigned by lottery (e.g. a coin flip decides which schools get extra funding). Nothing about school i, observed (controls) or unobserved (u_i), affects whether it is treated.

Random Assignment

$$\big(Y_i(1), Y_i(0)\big) \perp\!\!\!\perp X_{1i} \;\Rightarrow\; \mathbb{E}[Y_i(0) \mid X_{1i} = 1] = \mathbb{E}[Y_i(0) \mid X_{1i} = 0]$$

Selection bias $= 0$, so the naive comparison is causal:
$\mathbb{E}[Y_i \mid X_{1i} = 1] - \mathbb{E}[Y_i \mid X_{1i} = 0] = \text{ATT} = \text{ATE}$.



No dashed arrows: X_{1i} is unrelated to controls and u_i.

Regression $Y_i = \beta_0 + \beta_1 X_{1i} + u_i$ gives $\hat{\beta}_1 = \bar{Y}_{X_1=1} - \bar{Y}_{X_1=0}$; controls are optional (precision only).
Benchmark: each quasi-experimental design seeks a comparison where selection bias is (credibly) zero.

MLR and Conditional Mean Independence

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$$

Exogeneity (MLR.4) $\leftrightarrow$ Selection on Observables

$$\mathbb{E}[u_i \mid X_{1i}, \ldots, X_{ki}] = 0 \qquad \leftrightarrow \qquad \big(Y_i(1), Y_i(0)\big) \perp\!\!\!\perp X_{1i} \mid X_{2i}, \ldots, X_{ki}$$

"Once we compare units with the same controls, treatment is as good as random."

- $\hat{\beta}_j$ *always* estimates a partial correlation.
- It is a causal effect *only if* this assumption holds, and the assumption can never be tested directly: we cannot check that no relevant variable is missing.

Controls Can Change Everything: School Budget Example

	Dependent variable: `math4`	
	(1)	(2)
Expenditure per student ($1000)	−0.570	2.083***
	(0.428)	(0.350)
Share eligible for free lunch		−0.471***
		(0.014)
Observations	1,823	1,823
R^2	0.001	0.368

Adding one control flips the sign. Is column (2) causal? Only if nothing else in u_i correlates with spending, *conditional on lunch*. Can we be sure?

What Variation Identifies $\hat{\beta}_1$? Frisch–Waugh

$$Y_i = \beta_0 + \beta_1 X_{1i} + \boldsymbol{X}_{2i}'\boldsymbol{\beta}_2 + u_i$$

1. Regress X_{1i} on $\boldsymbol{X}_{2i}$, keep residuals $\tilde{X}_{1i}$
2. Regress Y_i on $\boldsymbol{X}_{2i}$, keep residuals $\tilde{Y}_i$
3. Regress $\tilde{Y}_i$ on $\tilde{X}_{1i} \Rightarrow$ slope $= \hat{\beta}_1$ from the full MLR

- $\hat{\beta}_1$ uses only the variation in X_1 that the controls do *not* explain.
- **Bridge:** panel fixed effects are Frisch–Waugh with unit dummies as $\boldsymbol{X}_2$. Partialling out the dummies is the same as demeaning each unit's data (the "within" transformation).

Interpreting Coefficients: The Building Blocks

Dummies and interactions

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 X_i + \beta_3 (D_i \times X_i) + u_i$$

Slope of X: β_2 if $D_i = 0$; $\beta_2 + \beta_3$ if $D_i = 1$. Jump at $X_i = 0$: β_1.

Logs

Model	Interpretation of β
level–level	$\Delta Y = \beta\, \Delta X$
level–log	$\Delta Y = (\beta/100)\, \%\Delta X$
log–level	$\%\Delta Y \approx (100\beta)\, \Delta X$
log–log	$\%\Delta Y = \beta\, \%\Delta X$

Where they reappear

- DiD: $\text{Treat}_i \times \text{Post}_t$
- Panel FE: one dummy per unit
- RDD: cutoff dummy $\times$ running variable
- Event study: one dummy per period relative to treatment
- Log outcomes $\Rightarrow$ effects in %

The Master Formula

Simple regression $Y_i = \beta_0 + \beta_1 X_i + u_i$. Under MLR.1–MLR.3:

Probability Limit of OLS

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \frac{\mathrm{Cov}(X_i, u_i)}{\mathrm{Var}(X_i)}$$

- Exogeneity $\Rightarrow \mathrm{Cov}(X_i, u_i) = 0 \Rightarrow$ OLS is consistent for the causal β_1.
- Endogeneity $\Rightarrow$ OLS is inconsistent: more data does *not* help.
- The sign of $\mathrm{Cov}(X_i, u_i)$ gives the direction of the bias.

Three common reasons why $\mathrm{Cov}(X_i, u_i) \neq 0$: **omitted variables**, **measurement error**, **reverse causality**.

Omitted Variable Bias

True model: $\log(wage_i) = \beta_0 + \beta_1 educ_i + \beta_2 ability_i + u_i$
Estimated: $\log(wage_i) = \beta_0 + \beta_1 educ_i + v_i, \quad v_i = \beta_2\, ability_i + u_i$

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \beta_2 \cdot \frac{\mathrm{Cov}(educ_i, ability_i)}{\mathrm{Var}(educ_i)}$$

	$\mathrm{Cov}(X_1, X_2) > 0$	$\mathrm{Cov}(X_1, X_2) < 0$
$\beta_2 > 0$	upward bias	downward bias
$\beta_2 < 0$	downward bias	upward bias

Ability raises wages and correlates with schooling $\Rightarrow$ OLS overstates the return to education.

Bridge: if the omitted variable is fixed over time (ability), panel FE and DiD remove it.

Measurement Error and Reverse Causality

Classical measurement error in X: observe $\widetilde{educ}_i = educ_i + w_i$, with w_i pure noise.

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 \cdot \underbrace{\frac{\mathrm{Var}(educ_i)}{\mathrm{Var}(educ_i) + \mathrm{Var}(w_i)}}_{\in(0,1]} \quad \Rightarrow \text{ attenuation bias towards zero}$$

Reverse causality (simultaneity):

$$grades_i = \beta_0 + \beta_1 \, funding_i + u_i$$
$$funding_i = \delta_0 + \delta_1 \, grades_i + v_i$$

$funding_i$ depends on $grades_i$, which depends on $u_i \Rightarrow \mathrm{Cov}(funding_i, u_i) \neq 0$, even if $\mathrm{Cov}(u_i, v_i) = 0$.

Bridge: adding controls fixes neither problem. Both are classic motivations for IV.

The Limits of "Just Add Controls"

- Adding every relevant covariate would remove the bias, but we can never verify that we have them all.
- Some confounders are unobservable by nature (ability, motivation, local politics).
- Controls do not solve measurement error or reverse causality.
- "Bad controls": controlling for a variable that is itself an outcome of treatment introduces bias.

Modern approach: instead of trying to observe every confounder, find a *source of variation* in X that is plausibly unrelated to the confounders.

That source of variation is the *research design*.

Sampling Distribution of $\hat{\beta}$

The **estimator** $\hat{\beta}$ is a random variable: each sample gives a different **estimate**.

- **Unbiased:** $\mathbb{E}[\hat{\beta}] = \beta$ (holds in any sample size)
- **Consistent:** $\hat{\beta} \xrightarrow{p} \beta$ as $n \to \infty$
- **Asymptotically normal:** $\hat{\beta} \overset{a}{\sim} N\Big(\beta,\ \mathrm{Var}(\hat{\beta} \mid \boldsymbol{X})\Big)$

OLS under MLR.1–MLR.4 has all three.
Bridge: the IV estimator is consistent but *biased* in finite samples (towards OLS), and the bias is severe when the instrument is weak.

The Sandwich and Robust Standard Errors

$$\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}) = (\boldsymbol{X}'\boldsymbol{X})^{-1}\, \boldsymbol{X}'\, \mathbb{E}[\boldsymbol{u}\boldsymbol{u}' \mid \boldsymbol{X}]\, \boldsymbol{X}\, (\boldsymbol{X}'\boldsymbol{X})^{-1}$$

Everything depends on the middle term, $\boldsymbol{\Omega} = \mathbb{E}[\boldsymbol{u}\boldsymbol{u}' \mid \boldsymbol{X}]$:

Assumption on $\boldsymbol{\Omega}$	Estimator	Status
$\sigma_u^2 \boldsymbol{I}_n$ (homoskedastic)	default `lm()` SEs	rarely credible
$\mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$	HC robust: $\hat{\sigma}_i^2 = \hat{u}_i^2$	cross-section default
block-diagonal by cluster	cluster-robust	panel / DiD default

Variance drivers (homoskedastic case): $\mathrm{Var}(\hat{\beta}_j \mid \boldsymbol{X}) = \dfrac{1}{n} \cdot \dfrac{\sigma_u^2}{\sigma_{X_j}^2} \cdot \dfrac{1}{1 - R_j^2}$

Bridge: FE and IV use less variation in X_j, so their standard errors are larger.

Clustered Standard Errors: Why

MLR.3 (random sampling) implies $\mathbb{E}[u_i u_j \mid \boldsymbol{X}] = 0$ for $i \neq j$. This fails when observations share shocks:

- Students in the same school; firms in the same region
- The same unit observed over time (panel): serially correlated u_{it}
- Treatment assigned at group level (a policy in some states)

Moulton factor (equal-sized groups of size $\bar{n}$):

$$\frac{\mathrm{Var}_{\mathsf{true}}(\hat{\beta})}{\mathrm{Var}_{\mathsf{iid}}(\hat{\beta})} \approx 1 + (\bar{n} - 1)\, \rho_X\, \rho_u$$

ρ_X, ρ_u: within-cluster correlation of regressor and error. If X is constant within clusters ($\rho_X = 1$) and groups are large, iid SEs can be wildly too small.

Clustered Standard Errors: How

Allow arbitrary correlation *within* clusters $g = 1, \ldots, G$, independence *across* them:

$$\widehat{\mathrm{Var}}_{CL}(\hat{\boldsymbol{\beta}}) = (\boldsymbol{X}'\boldsymbol{X})^{-1}\Big(\sum_{g=1}^{G} \boldsymbol{X}_g' \, \hat{\boldsymbol{u}}_g \hat{\boldsymbol{u}}_g' \, \boldsymbol{X}_g\Big)(\boldsymbol{X}'\boldsymbol{X})^{-1}$$

(in practice multiplied by a small-sample correction $\frac{G}{G-1}\frac{n-1}{n-k-1}$)

Practical rules

- Cluster at the level at which treatment is assigned (Abadie et al., 2023).
- Panel/DiD: cluster by unit to allow serial correlation (Bertrand, Duflo & Mullainathan, 2004).
- Asymptotics are in G, not n: with few clusters ($G \lesssim$ 30–50) use the wild cluster bootstrap.

```
library(fixest)
feols(y ~ x | unit + year, data = df, cluster = ~unit)
```

Tests and Confidence Intervals: Correct Interpretation

t-statistic: $T = \dfrac{\hat{\beta}_j - \beta_{j,0}}{\widehat{SE}(\hat{\beta}_j)} \overset{a}{\sim} N(0,1)$ under H_0. Reject at level α if $|T| > z_{1-\alpha/2}$.

- **p-value:** the probability, *if H_0 were true*, of a test statistic at least as extreme as the one observed. It is **not** the probability that H_0 is true.
- **95% CI** $[\hat{\beta}_j \pm 1.96\,\widehat{SE}(\hat{\beta}_j)]$: across repeated samples, 95% of intervals constructed this way cover β_j. A given realized interval either contains β_j or it does not.
- **Statistical vs. economic significance:** report effect sizes (e.g. the effect of a 1-SD change in X_j), not just stars.

The SEs (robust or clustered) determine everything on this slide, so choosing them is a substantive decision.

One Map: What Each Design Assumes Instead

Design	Replaces $\mathbb{E}[u_i \mid X_i] = 0$ with	Estimand	Inference
RCT	Random assignment	ATE	robust
MLR	Selection on observables	ATE (if CIA)	robust
IV	Instrument exogenous + relevant	LATE	robust / clustered
Panel FE	Strict exogeneity given unit FE	within-unit effect	cluster by unit
DiD	Parallel trends	ATT	cluster by unit/group
RDD	Continuity at the cutoff	local effect at cutoff	robust / bias-corrected

Each design is still a regression. What differs is *which variation* identifies β, and *why* that variation is credibly unrelated to u_i.

Instrumental Variables

Instrument Conditions (endogenous X_i, instrument Z_i)

1. **Relevance:** $\mathrm{Cov}(Z_i, X_i) \neq 0$ (testable: first stage)
2. **Exogeneity / exclusion:** $\mathrm{Cov}(Z_i, u_i) = 0$; Z affects Y only through X (not testable)

$$\hat{\beta}_1^{IV} = \frac{\widehat{\mathrm{Cov}}(Z_i, Y_i)}{\widehat{\mathrm{Cov}}(Z_i, X_i)} = \frac{\text{reduced form}}{\text{first stage}} \qquad \text{(matrix: } \hat{\boldsymbol{\beta}}^{IV} = (\boldsymbol{Z}'\boldsymbol{X})^{-1}\boldsymbol{Z}'\boldsymbol{Y}\text{)}$$

- Compare with OLS $(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y}$: same structure, with $\boldsymbol{Z}$ supplying the exogenous variation.
- 2SLS: regress X on Z (and controls), then Y on $\hat{X}$. Use the IV routine; SEs from a manual second stage are wrong.
- Weak instruments: report first-stage F. The rule of thumb $F > 10$ is now considered too lenient for t-based inference (Lee et al., 2022).

IV with Heterogeneous Effects: LATE

Binary Z_i and X_i. Potential treatments $X_i(1), X_i(0)$ define four types:

	$X_i(0) = 0$	$X_i(0) = 1$
$X_i(1) = 1$	complier	always-taker
$X_i(1) = 0$	defier	never-taker

With independence, exclusion, relevance and **monotonicity** (no defiers):

$$\frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[X_i \mid Z_i = 1] - \mathbb{E}[X_i \mid Z_i = 0]} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{complier}] = \text{LATE}$$

IV recovers the effect only for units whose behavior the instrument changes. Different instruments $\Rightarrow$ different complier populations $\Rightarrow$ different LATEs.

Panel Data and Fixed Effects

$$Y_{it} = \beta_1 X_{it} + \alpha_i + \lambda_t + u_{it}$$

- α_i: everything about unit i that is fixed over time (ability, geography, institutions), observed or not.
- λ_t: common shocks in period t. Estimate via dummies or demeaning $\ddot{Y}_{it} = Y_{it} - \bar{Y}_i$ (Frisch–Waugh).

What carries over from the review

- OVB: time-invariant confounders are eliminated; time-varying ones are not.
- Perfect multicollinearity: effects of time-invariant regressors (e.g. gender) cannot be estimated.
- Variance: only within-unit variation is used $\Rightarrow$ larger SEs; attenuation from measurement error is often worse.
- Inference: u_{it} is serially correlated $\Rightarrow$ cluster by unit.

Difference-in-Differences

$$Y_{it} = \gamma_0 + \gamma_1 \text{Treat}_i + \gamma_2 \text{Post}_t + \delta\,(\text{Treat}_i \times \text{Post}_t) + u_{it}$$

	Pre	Post
Treated	$\gamma_0 + \gamma_1$	$\gamma_0 + \gamma_1 + \gamma_2 + \delta$
Control	γ_0	$\gamma_0 + \gamma_2$

$$\delta = (\bar{Y}_{T,post} - \bar{Y}_{T,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre})$$

δ: the interaction coefficient from the building-blocks slide.

Parallel Trends

Absent treatment, treated and control units would have followed the same path:

$$\mathbb{E}[Y_{it}(0) - Y_{i,t-1}(0) \mid \text{Treat}_i = 1] = \mathbb{E}[Y_{it}(0) - Y_{i,t-1}(0) \mid \text{Treat}_i = 0]$$

- Identifies the **ATT**. Level differences between groups are allowed; differences in trends are not.
- Cluster at the level at which treatment is assigned.

Event Studies and Staggered Adoption

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \delta_k \, 1[t - E_i = k] + u_{it}$$

- One dummy per period relative to treatment date E_i; $k = -1$ is omitted (perfect multicollinearity, otherwise).
- Plot $\hat{\delta}_k$ with 95% CIs (the coefficient plot from the inference lecture).
- Leads ($k < -1$) near zero: consistent with parallel pre-trends. This supports the assumption but does not prove it.

Caveat: when units adopt treatment at different times and effects vary over time, two-way FE compares later-treated units with already-treated ones and can even flip signs (Goodman-Bacon, 2021). Use heterogeneity-robust estimators: Callaway & Sant'Anna (2021), Sun & Abraham (2021), de Chaisemartin & D'Haultfœuille (2020).

Regression Discontinuity Design

Treatment switches at a cutoff c of a running variable R_i: $X_i = 1[R_i \geq c]$.

$$Y_i = \beta_0 + \tau X_i + \beta_1(R_i - c) + \beta_2 X_i \times (R_i - c) + u_i, \qquad |R_i - c| \leq h$$

- τ: jump in $\mathbb{E}[Y \mid R]$ at the cutoff. Separate slopes on each side via the interaction.
- **Continuity:** $\mathbb{E}[Y_i(0) \mid R_i = r]$ and $\mathbb{E}[Y_i(1) \mid R_i = r]$ are continuous at c. Units just above and just below are comparable.
- Estimand: effect *at the cutoff*. External validity away from c is limited.

Practice

- Local linear fits with data-driven bandwidth h and bias-corrected robust CIs (`rdrobust`). Avoid high-order global polynomials (Gelman & Imbens, 2019).
- Check for manipulation of R (density test) and balance of covariates at c.

Recap

- Causal questions come first: potential outcomes define *what* we estimate (ATE, ATT, LATE).
- OLS estimates partial correlations. It is causal only under $\mathbb{E}[u_i \mid X_i] = 0$, which is unverifiable with observational data.
- One formula diagnoses the failure: $\operatorname{plim} \hat{\beta}_1 = \beta_1 + \operatorname{Cov}(X, u)/\operatorname{Var}(X)$.
- Inference depends on $\boldsymbol{\Omega}$: robust SEs for cross-sections, clustered SEs whenever errors are correlated within groups.
- IV, FE, DiD and RDD reuse the same tools (dummies, interactions, Frisch–Waugh, the sandwich) with a different identifying assumption. The quality of a design is the credibility of that assumption.

Recommended Literature

- Angrist & Pischke (2015), *Mastering 'Metrics*: Ch. 1 (RCT, selection bias), 3 (IV), 4 (RDD), 5 (DiD)
- Angrist & Pischke (2009), *Mostly Harmless Econometrics*: Ch. 8 (clustering)
- Cunningham (2021), *Causal Inference: The Mixtape*: Ch. 4 (potential outcomes)
- Huntington-Klein (2021), *The Effect*
- Heiß, *Using R for Introductory Econometrics*: implementation

Abadie, Athey, Imbens & Wooldridge (2023, QJE); Bertrand, Duflo & Mullainathan (2004, QJE); Goodman-Bacon (2021, JoE); Callaway & Sant'Anna (2021, JoE); Sun & Abraham (2021, JoE); de Chaisemartin & D'Haultfœuille (2020, AER); Gelman & Imbens (2019, JBES); Lee, McCrary, Moreira & Porter (2022, AER).